

PRANVEER SINGH INSTITUTE OF TECHNOLOGY

B.Tech Computer Science and Engineering — Semester Examination Simulation

SUBJECT: DESIGN ANALYSIS AND ALGORITHM

Paper Set:	Set 02	Duration:	3 Hours
Maximum Marks:	100 Marks	Date:	September 2026

SECTION A

Answer all questions in brief. (10 x 2 = 20 Marks)

- Q1(a).** Write down the basic properties or conditions required for Complexity of Algorithms. [2M]
- Q1(b).** Differentiate in brief between Comparison of Sorting Algorithms and Growth of Functions. [2M]
- Q1(c).** List any two advantages and limitations of Growth of Functions. [2M]
- Q1(d).** Give a real-world example illustrating Performance Measurements. [2M]
- Q1(e).** Define Algorithms and state its key application in computer systems. [2M]
- Q1(f).** What is the primary significance of Analyzing Algorithms? Explain in brief. [2M]
- Q1(g).** State the fundamental principles governing Sorting in Linear Time. [2M]
- Q1(h).** Discuss the key methodologies and implementations of Complexity of Algorithms. [2M]
- Q1(i).** Discuss the key methodologies and implementations of Comparison of Sorting Algorithms. [2M]
- Q1(j).** Discuss the key methodologies and implementations of Growth of Functions. [2M]

SECTION B

Attempt any three of the following questions. (3 x 10 = 30 Marks)

- Q2.** Design a solution using Performance Measurements for an enterprise system and trace each step. [10M]
- Q3.** Explain the theoretical foundation of Algorithms and prove its correctness. [10M]
- Q4.** Write a step-by-step algorithm or procedure for Analyzing Algorithms with illustrative inputs. [10M]
- Q5.** Discuss the various challenges encountered in Sorting in Linear Time and how they are mitigated. [10M]
- Q6.** Explain Complexity of Algorithms in detail with suitable architectural diagrams and examples. [10M]

SECTION C

Attempt any five questions from this section. (5 x 10 = 50 Marks)

- Q5.** Design a solution using Comparison of Sorting Algorithms for an enterprise system and trace each step. [10M]
- Q6.** Explain the theoretical foundation of Growth of Functions and prove its correctness. [10M]

- Q7.** Write a step-by-step algorithm or procedure for Performance Measurements with illustrative inputs. [10M]
- Q8.** Discuss the various challenges encountered in Algorithms and how they are mitigated. [10M]
- Q9.** Explain Analyzing Algorithms in detail with suitable architectural diagrams and examples. [10M]
- Q10.** Discuss the algorithmic approach for Sorting in Linear Time. Analyze its time and space complexity. [10M]
- Q11.** Compare and contrast Complexity of Algorithms with Comparison of Sorting Algorithms. Highlight performance trade-offs. [10M]